Sangwoo HAN

Undergraduate Researcher
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Personal Website
Google Scholar
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Objective

Senior undergraduate in Artificial Intelligence Software at Cheongju University, seeking a research position for M.S./Ph.D. admission in Spring 2027. Broadly interested in robust embodied AI systems, with a core focus on robot learning (RL/IL) and Vision-Language-Action (VLA) models for manipulation, emphasizing effective Sim2Real transfer. Additionally, interested in sensor fusion, residual reinforcement learning (Residual RL), and offline reinforcement learning (Offline RL) for building robust and adaptive control and perception in robotic agents.

Education

2025–present **B.S. Candidate, Artificial Intelligence Software**, Cheongju University, Republic of Korea.
GPA: 4.38/4.50 Expected graduation: Feb. 2027

Research Interests

- Robot foundation models for dexterous manipulation and embodied intelligence
- Reinforcement learning and imitation learning for robot decision-making and skill acquisition
- Sim-to-Real transfer
- Sensor fusion, residual reinforcement learning (Residual RL), and offline reinforcement learning (Offline RL) for robust and adaptive robot control

Skills

Programming	Python (Proficient), C++ (Familiar), JavaScript (Familiar)
Robot Platforms	Rainbow Robotics RB5, Elephant Robotics myCobot 630 Pro
End-effectors & Sensors	Robotiq 2F-85, myCobot Gripper F100, myCobot Air Compressor (pneumatic), AFT-200 F/T sensor, Intel RealSense
Robotics	ROS2, Nav2, MoveIt 2, Gazebo, Webots, Isaac Sim, Drake
AI/ML	PyTorch, JAX
Tools	Git, Linux (Ubuntu/WSL2), Docker
Languages	Korean (Native), English (Intermediate), Japanese (Intermediate), German (Beginner)

Research Experience

Nov. 2025–Dec. 2025 **Real-time Physical Document Digitization for Video Conferencing**
Industry-Academic Project with Gooroome
- Built pre-/post-processing OCR pipelines for low-resolution webcam documents, reducing Levenshtein edit distance by 3.2× over baseline.

Nov. 2025–Feb. 2026 **Development of High-Performance OCR System for Industrial LOT Cards**
Industry-Academic Project with International Electric Co. & RISE Chungbuk AI Center
- Developed a robust OCR-to-table pipeline handling blur, glare, and skew in industrial LOT cards.
- Implemented layout-based region clustering, re-OCR, and rule-based validation to ensure reliable CSV/XLSX export.

Jul. 2026–Jan. 2027 **Development of AI Vision-Based Automatic Pattern and Defect Detection and Analysis Technology for Wafer Master Substrates**
Industry-Academic Project with KAPS & RISE Chungbuk AI Center

Publications & Manuscripts

Publication notation	
†	Corresponding author.
Sangwoo HAN	The CV author's name is shown in bold.

Peer-Reviewed Journal Articles

J1. **Sangwoo HAN**, and H. Choi†, "ROEP: A Robotics-Oriented Evaluation Protocol for Deployment-Facing Vision-Language-Action Manipulation Policies," *MDPI Sensors*, vol. 26, no. 15, p. 4757, 2026.
We propose a robotics-oriented evaluation protocol (ROEP) designed to systematically assess deployment-facing Vision-Language-Action (VLA) manipulation policies under real-world perturbations and operational conditions.

Submitted Manuscripts (Under Review)

S2. J. Lee, Y. Cheon, **Sangwoo HAN**, C. Yu, H. Lee, J. Lee, and Y. Lee†, "PIVOT: Part-Wise Identification of Volume-Aware Density for Articulated Objects," submitted to *IEEE International Conference on Robotics and Automation (ICRA 2027)*, 2026.

S1. **Sangwoo HAN**, and H. Choi†, "ReGuide: Reliability-Guided Source Selection for Robot Demonstration Generation," submitted to *Conference on Robot Learning (CoRL 2026) Workshop*, 2026.

Test Scores

English	TOEIC 920 (out of 990)

Activities and Research Involvement

Undergraduate Student Researcher	VICA Lab (advised by Prof. Hyunguk Choi), Cheongju University
Summer Research Intern	ROBIN Lab (advised by Prof. Yongseok Lee), DGIST
Member	CPR (AI Software Engineering Student Club), Cheongju University